MLF Week 3 : Linear Algebra - Least Squares Regression

Lecture 1 : Four Fundamental Subspaces

Linear Algebra

Four fundamental subspaces of a given matrix A

Column Space $C(A)$:

$$A = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$$

$C(A) = \text{span}(u_1, \ldots, u_n) = \{ \text{linear combinations of the vectors } u_1, \ldots u_n \}$

Solving $Ax=b$:

For what b does $Ax=b$ have a solution?
$b \in C(A)$

Let's look at an example

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

$Ax=b$ is solvable for $b \in C(A)$.

Does every vector in $\mathbb{R}^4$ belong to $C(A)$?

Some vectors in $\mathbb{R}^4$ are not in $C(A)$, because $Ax=b$ is "4 equations in 3 unknowns"

For the example above, $\text{col}3 = \text{col}1 + \text{col}2$

So, $C(A) = \text{span}(\text{col}1, \text{col}2)$

$\Rightarrow C(A)$ is a two-dimensional subspace of $\mathbb{R}^4$

Null space $N(A)$

$$N(A) = \{ x \mid Ax = 0 \}$$

Why is $N(A)$ a subspace? $x_1, x_2 \in N(A)$

Then $Ax_1 = 0$, $Ax_2 = 0$

$\Rightarrow A(x_1 + x_2) = 0$

$\Rightarrow x_1 + x_2 \in N(A)$

$x \in N(A)$, α is a scalar

$A(\alpha x) = \alpha Ax$

$= 0$

So, $\alpha x \in N(A)$

Hence, $N(A)$ is a subspace.

Example: $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$

Find an x s.t. $Ax = 0$

$\Leftrightarrow$ a linear combination of the columns of A should result in the zero vector.

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in N(A)$$

So, $N(A)$ is a line in $\mathbb{R}^3$

Remark 1: If A is invertible, then $N(A)$ has "zero" only,

and $C(A)$ is the whole space.

In this case, $Ax = b$ has a unique solution $x = A^{-1}b$

Else, $N(A)$ has $x_n \neq 0$ and $Ax = b$ solutions are of the form $x = x_p + x_n$

$Ax_p = b$, $Ax_n = 0$

Remark 2: Can we Gaussian elimination to find the Null space of a matrix A i.e., solve $Ax=0$.

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \longrightarrow \underbrace{\begin{bmatrix} \boxed{1} & 2 & 2 & 2 \\ 0 & 0 & \boxed{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Call this } U}$$

pivots

$Ux=0$

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

col 1, col 3 → pivot cols, col 2, col 4 → free variables

Set $x_2 = 1,\ x_4 = 0 \Rightarrow 2x_3 + 4x_4 = 0 \Rightarrow x_3 = 0$

$x_1 + 2x_2 + 2x_3 + 2x_4 = 0 \Rightarrow x_1 = -2$

$$u = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \in N(A)$$

Set $x_2 = 0,\ x_4 = 1 \Rightarrow 2x_3 + 4x_4 = 0 \Rightarrow x_3 = -2$

$x_1 + 2x_2 + 2x_3 + 2x_4 = 0$

$\Rightarrow x_1 - 4 + 2 = 0 \Rightarrow x_1 = 2$

$$v = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \in N(A)$$

$N(A) = \text{span}(u, v)$

Rank = number of pivot columns. In the example above, rank(A) = 2

Nullity = number of free variables. In the example above, nullity(A) = 2

Rank = dim(C(A)) , Nullity = dim(N(A)).

If A has n columns, then $\boxed{\text{rank} + \text{nullity} = n}$ $\Leftrightarrow$ rank(A) = r, nullity(A) = n − r

Row space R(A): column space of A^T $\Leftrightarrow$ span of rows of A

$R(A) = C(A^T)$

Important fact: col rank = dim(C(A)), row-rank = dim(R(A))

col rank = row rank

So far, we looked at $C(A)$, $N(A)$, $R(A)$ $(= C(A^T))$

Left Null space $N(A^T) = \{ y \mid A^T y = 0 \} = \{ y \mid y^T A = 0 \}$

For a m×n matrix A, $\underbrace{\begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix} \begin{bmatrix} & & \\ & A & \\ & & \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}}_{\text{a linear combination of rows leading to zero vector}}$

Remark: ① A is a m×n matrix

$\dim(C(A)) + \dim(N(A))$ = number of columns of A = n

"r" + "n−r" = n

② $\dim C(A^T) + \dim(N(A^T))$ = number of rows = m

$\dim(C(A^T)) = r \Rightarrow$ $r + \dim(N(A^T)) = m \Rightarrow \dim(N(A^T)) = m - r$

Example:

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

want $\quad y^T A = 0$

$$\begin{bmatrix} 1 & 1 & -1 \end{bmatrix} \in N(A^T)$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \qquad m = n = 2$$

$C(A)$ = line through $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ $\Rightarrow$ $\text{rank}(A) = r = 1$

$N(A)$ = line through $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ $\Rightarrow$ $\text{nullity}(A) = n - r = 1$

$R(A) = C(A^T)$ = line through $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\quad \dim(R(A)) = r = 1$

Left Null space $N(A^T)$ = line through $\begin{bmatrix} -3 \\ 1 \end{bmatrix}$ $\quad \dim(N(A^T)) = 1$

Homework: Work out the four fundamental spaces $C(A)$, $N(A)$, $C(A^T)$, $N(A^T)$ for

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

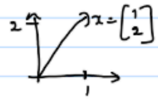
Lecture 2 : Orthogonal Vectors and Subspaces

Least squares

Background on orthogonality

We will look at
1. Orthogonal vectors, length of a vector
2. Projections
3. Least-squares regression

Length of a vector



$$\|x\|^2 = x_1^2 + x_2^2$$

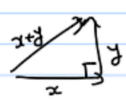
$$\left\| \begin{bmatrix}1\\2\end{bmatrix} \right\|^2 = 1^2 + 2^2 = 5$$

In general, for $x = \begin{bmatrix}x_1\\ \vdots \\ x_n\end{bmatrix} \in \mathbb{R}^n$, $\|x\|^2 = x_1^2 + x_2^2 - - - + x_n^2$

Orthogonal vectors: $x \perp y$ if $\underbrace{x^T y}_{\text{Inner product}} = 0$

$x = \begin{bmatrix}x_1\\ \vdots \\ x_n\end{bmatrix}$, $y = \begin{bmatrix}y_1\\ \vdots \\ y_n\end{bmatrix}$ Then $x^T y = \sum_{i=1}^{n} x_i y_i$

From Pythagoras theorem:



$$\|x\|^2 + \|y\|^2 = \|x+y\|^2$$
$$x^T x + y^T y = (x+y)^T (x+y)$$
$$\cancel{x^T x} + \cancel{y^T y} = \cancel{x^T x} + \cancel{y^T y} + x^T y + y^T x$$
$$2\, x^T y = 0 \Rightarrow \boxed{x^T y = 0} \rightarrow \text{orthogonal to } y.$$

x is

Remark: 1. 0 is orthogonal to every x

② If $\{v_1 \dots v_k\}$ are mutually orthogonal "non-trivial" set of vectors
$\{v_1 \dots v_k\}$ is a linearly independent set

Why? Suppose $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$

$\Rightarrow v_1^T (c_1 v_1 + \dots + c_k v_k) = 0$

$\Rightarrow c_1 \, v_1^T v_1 = 0$, but $\|v_1\| \neq 0 \Rightarrow c_1 = 0$

Similarly $c_i \neq 0 \ \forall i$

Examples: ① $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

② $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

② If $\{v_1 \dots v_k\}$ are mutually orthogonal "non-trivial" set of vectors
$\{v_1 \dots v_k\}$ is a linearly independent set

Why? Suppose $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0$

$\Rightarrow v_1^T (c_1 v_1 + \dots + c_k v_k) = 0$

$\Rightarrow c_1 \, v_1^T v_1 = 0$, but $\|v_1\| \neq 0 \Rightarrow c_1 = 0$

Similarly $c_i \neq 0 \ \forall i$

Correction : Similarly C_i not equal to 0 should be replaced by Similarly C_i=0

Orthonormal vectors: $\{u, v\}$ orthonormal if $\underbrace{u^T v = 0}_{\text{orthogonal}}$ and $\underbrace{\|u\| = \|v\| = 1}_{\text{unit length}}$

Example: $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, $\left\{ \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \right\}$

Orthogonal subspaces:

U, V are orthogonal subspaces if

$x^T y = 0 \quad \forall x \in U, y \in V$

Examples: 1. $\{0\} \perp$ subspace

2. $U = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$ $\quad V = \left\{ \begin{pmatrix} 0 \\ 0 \\ 2 \\ 1 \end{pmatrix} \right\}$

$U \perp V$

$W = \text{span}\left\{ \begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\}$

Correction : W=span{(0,0,-2,1)} should be replaced by W=span{(0,0,-1,2)}

Then $W \perp U$, $W \perp V$

$$W = \text{Span}\left\{ \begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\}$$

Then $W \perp U, \quad W \perp V$

Orthogonality wrt four fundamental subspaces of a matrix A:

Claim: (1) $R(A) \perp N(A)$

Proof: $x \in N(A)$

$Ax = 0 \quad (A \text{ is } m \times n)$

$$\begin{bmatrix} - \text{row1} - \\ - \text{row2} - \\ \vdots \\ - \text{row}\, m - \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

The equation above implies
row1 $\perp x$
row 2 $\perp x$
$\vdots$
row m $\perp x$

Thus, any linear combination $\big(\underbrace{c_1\, \text{row1} + c_2\, \text{row2} + \cdots + c_m\, \text{row}\,m}_{\in R(A)}\big) \perp x$

So, $R(A) \perp N(A)$

$\Leftrightarrow C(A^T) \perp N(A)$

Claim 2: $C(A) \perp N(A^T)$

"follows from claim 1".

Example $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$

$\text{rank}(A) = \dim(C(A)) = 1$

$x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \in N(A)$ since "col 2 = 2 × col 1"

Row-space $R(A)$ = line through $\begin{bmatrix} 1 & 2 \end{bmatrix}$

Null space $N(A)$ = line through $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = 0$ verifies $R(A) \perp N(A)$

$C(A)$ = line through $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$N(A^T) = y_1 + 2y_2 + 3y_3 = 0$ ← is a plane

$y^T A = 0$

Connect this to the claim $C(A) \perp N(A^T)$

A dimension check:

$\dim C(C(A)) + \dim(N(A)) = 2$

$1 + 1$ number of cols

$\dim C(A^T) + \dim(N(A^T)) = 3$

$1 + 2 =$ number of cols

QN - 4,2

Lecture 3 : Projections

Projections and least squares

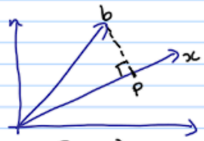


Fig (1)

Want to project b onto the line through x, or, more generally onto the column space of a matrix A.

① Why project?

② Given a basis for a subspace S (e.g., spanned by cols of A), is there an easy to calculate the projection p of b onto S?

On ①: Suppose we are given $(x_1, b_1) \; .... \; (x_n, b_n)$

e.g. $\begin{matrix} 2x = b_1 \\ 3x = b_2 \\ 4x = b_3 \end{matrix}$ or $\begin{matrix} x + 2y = 4 \\ x + 3y = 5 \\ 2x + 4y = 6 \end{matrix}$

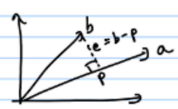
"These are inconsistent" $\Rightarrow$ No solution that satisfies this system of equations

Matrix view: $Ax = b$

Inconsistent if $b \notin C(A)$

In such situations, it makes sense to project b onto $C(A)$.

Projection onto a line:



$$p = \hat{x} a$$

$$e = b - p = b - \hat{x} a$$

$$e \perp a$$

$$(b - \hat{x} a) \perp a$$

$$a^T (b - \hat{x} a) = 0 \quad \text{leading to} \quad \hat{x} = \frac{a^T b}{a^T a} \Rightarrow p = \hat{x} a = \left(\frac{a^T b}{a^T a}\right) a$$

Cauchy - Schwarz inequality:

$$\|e\|^2 = \|b - p\|^2 \geq 0$$

$$\left\| b - \frac{a^T b}{a^T a} a \right\|^2 = b^T b - 2\frac{(a^T b)^2}{a^T a} + \left(\frac{a^T b}{a^T a}\right)^2 a^T a$$

$$= \frac{(b^T b)(a^T a) - (a^T b)^2}{(a^T a)} \geq 0$$

$$\Rightarrow (b^T b)(a^T a) \geq (a^T b)^2$$

(or) $\boxed{|a^T b| \leq \|a\| \, \|b\|}$ → Cauchy - Schwarz inequality

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Projection matrix: Recall $p = \left(\frac{a^T b}{a^T a}\right) a = \left(\frac{a\, a^T}{a^T a}\right) b$

Let $\mathbb{P} = \frac{a a^T}{a^T a}$. Then, projection of b onto a is $\mathbb{P} b$

To project any vector b, just left multiply by the projection matrix '$\mathbb{P}$'.

Example: $a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Projection matrix is $\mathbb{P} = \frac{a a^T}{a^T a} = \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$

Observe that (i) $\mathbb{P}$ is symmetric

(ii) $\mathbb{P}^2 = \mathbb{P}$ i.e., $\mathbb{P}^2 b = \mathbb{P} b$ (idea: $\mathbb{P}b$ is already on the line through a. So, another round of projection won't change it)

(iii) Column space $C(\mathbb{P})$ = line through $a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Null space $N(\mathbb{P})$ = plane orthogonal to $a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Rank $r(P) = 1$

(iv) $a = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ $\mathbb{P} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$

check this

Lecture 4 : Least Squares and Projections onto a Subspace

Back to least squares

$$\left.\begin{aligned} 2x &= b_1 \\ 3x &= b_2 \\ 4x &= b_3 \end{aligned}\right\}$$ This system is solvable if b is on the line through $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

Suppose we have a vector "b" that leads to an "inconsistent" system.

We could pick a subset of equations & solve it exactly.

Problem with this approach: large errors in some inputs & no error in others

Reasonable alternative: minimize average error

$$E^2 = (2x-b_1)^2 + (3x-b_2)^2 + (4x-b_3)^2$$

Want to minimize the sum of squares E^2.

$$\frac{dE^2}{dx} = 0 \iff 2\left[2(2x-b_1) + 3(3x-b_2) + 4(4x-b_3)\right] = 0$$

leading to $\hat{x} = \dfrac{2b_1 + 3b_2 + 4b_3}{2^2 + 3^2 + 4^2} = \dfrac{a^T b}{a^T a}$ with $a = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

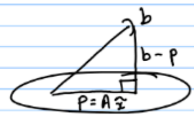
connects to projections

Bottom line: Taking derivative & finding the minima turns out to be the same as performing a projection.

Projection onto a subspace:

$Ax = b$, A is $m \times n$, $m > n$

Want: projection of b onto column space $C(A)$



S = span (columns of A)

Projection of b onto S is $p = A\hat{x}$

Orthogonal vector $e = b - p = b - A\hat{x}$

Q: How to find $\hat{x}$?

Observe $e \perp$ every vector in $C(A)$

Recall that $C(A) \perp N(A^T)$, i.e., $N(A^T)$ is the orthogonal complement of $C(A)$, i.e., every vector in $C(A)$ is orthogonal to every vector in $N(A^T)$ & any given vector is in either $C(A)$ or $N(A^T)$

Where does e belong? $e \in N(A^T) \Rightarrow A^T e = 0 \Rightarrow A^T(b - A\hat{x}) = 0$

leading to $\boxed{A^T A \hat{x} = A^T b} \longrightarrow$ equation to solve to obtain the projection of b onto $C(A)$.

(Note: Even if $Ax = b$ is not solvable, $A^T A \hat{x} = A^T b$ has a solution)

Alternative route to the above equation:

$$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{bmatrix} \qquad \left.\begin{matrix} a_1^T e = 0 \\ \vdots \\ a_n^T e = 0 \end{matrix}\right\} \Leftrightarrow \begin{matrix} a_1^T (b - A\hat{x}) = 0 \\ \vdots \\ a_n^T (b - A\hat{x}) = 0 \end{matrix}$$

$$\Updownarrow$$

$$\begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} \begin{bmatrix} b - A\hat{x} \end{bmatrix} = 0 \Leftrightarrow A^T(b - A\hat{x}) = 0$$

leading to $\boxed{A^T A \hat{x} = A^T b}$

Bottomline: Solving $A^T A \hat{x} = A^T b$ leads to an $\hat{x}$ that minimizes $\|Ax-b\|^2$

Connection of projections to least squares

Remarks:

① Suppose columns of A are linearly independent (l.i.)

Then, $\underbrace{A^T A}_{\text{square, symmetric}}$ is invertible (Why?)

Solving $A^T A \hat{x} = A^T b$ when $(A^T A)$ is invertible

$$\boxed{\hat{x} = (A^T A)^{-1} A^T b}$$

Projection $P = A\hat{x} = A(A^T A)^{-1} A^T b$

② $b \in C(A)$ i.e., $b = Ax$

$$p = A(A^T A)^{-1} A^T b = A \underbrace{(A^T A)^{-1} A^T A}_{= I} x = Ax = b$$

③ $b \in N(A^T)$

$$p = A(A^T A)^{-1} A^T b = 0 \quad \text{since } A^T b = 0$$

④ A is square & invertible $\Leftrightarrow$ $C(A) = \mathbb{R}^n$

$$p = A(A^T A)^{-1} A^T b = A A^{-1} (A^T)^{-1} A^T b = b$$

⑤ A is rank one i.e., $A = \begin{bmatrix} | \\ a \\ | \end{bmatrix}$

Then, $\hat{x} = \dfrac{a^T b}{a^T a}$ → coincides with what we derived earlier for projection onto a line.

Projection matrix $\mathbb{P} = A(A^TA)^{-1}A^T$

Symmetric $\mathbb{P}^T = \mathbb{P}$

$$\left(A(A^TA)^{-1}A^T\right)^T = (A^T)^T\left((A^TA)^{-1}\right)^T A^T$$
$$= A(A^TA)^{-1}A^T = \mathbb{P}$$

$\mathbb{P}^2 = \mathbb{P}$

$$\mathbb{P}^2 = A(A^TA)^{-1}A^T A(A^TA)^{-1}A^T$$
$$= A(A^TA)^{-1}A^T = \mathbb{P}$$

So, Projection matrix is symmetric & satisfies $\mathbb{P}^2 = \mathbb{P}$

The converse is also true.

<u>Claim:</u> If $\mathbb{P}^2 = \mathbb{P}$ & $\mathbb{P}$ is symmetric, then $\mathbb{P}$ is a projection matrix

$\mathbb{P}b$ = projection of b onto the column Space of $\mathbb{P}$

$$(b - \mathbb{P}b)^T \mathbb{P}c = b^T(I - \mathbb{P})^T \mathbb{P}c = b^T(\mathbb{P} - \mathbb{P}^2)c = 0 \text{ since } \mathbb{P}^2 = \mathbb{P}.$$

QN : 2

Lecture 5 : Example of Least Squares

Least Squares

Simple case : One dimension

Dataset: (x_1, b_1) - - - - - (x_m, b_m)

$$b_i = \theta' x_i + \theta'' \quad , \quad i = 1, \ldots, m \quad — \quad (1)$$

linear fit

offset

System of equations (1) is equivalent to

$$\begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix} \begin{bmatrix} \theta' \\ \theta'' \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

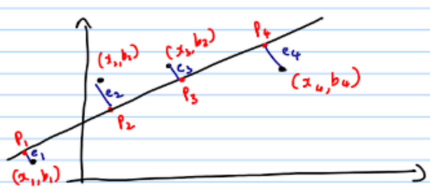
$A\theta = b$, where $\theta = \begin{pmatrix} \theta' \\ \theta'' \end{pmatrix}$

$A\theta = b$ may be inconsistent

Least squares approach: minimize $E^2 = \|b - A\theta\|^2$

$$= (b_1 - \theta' x_1 - \theta'')^2 + \cdots + (b_m - \theta' x_m - \theta'')^2$$

$$(\hat{\theta}', \hat{\theta}'') = \arg\min_{\theta} \|b - A\theta\|^2$$



If the points b_1, b_2, b_3, b_4 lie on a line, then $p_1 = b_1$, $p_2 = b_2$, $p_3 = b_3$, $p_4 = b_4$ & $E^2 = 0$ & $Ax = b$ can be solved

If not, then minimize $E^2 = \|A\theta - b\|^2$

Example:

$$A\,\theta = b$$

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \theta' \\ \theta'' \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

Is $A\theta = b$ consistent? Or does $b \in C(A)$?

$$\left[\begin{array}{cc|c} -1 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 2 & 2 \\ 0 & 3 & 5 \end{array}\right]$$

$$\downarrow$$

inconsistent $A\theta = b$ or $b \notin C(A)$ $\longleftarrow$ $\left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{array}\right] \longleftarrow \left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{array}\right]$

Least-squares:

$$A^T A\,\hat{\theta} = A^T b \quad , \text{ where } \quad \hat{\theta} = \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix}$$

$$A^T A = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}, \quad A^T b = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

$$A^T A\,\hat{\theta} = A^T b \iff \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

$$6\,\hat{\theta}' + 2\,\hat{\theta}'' = 6$$
$$2\,\hat{\theta}' + 3\,\hat{\theta}'' = 5$$

leads to $\quad \hat{\theta}'' = \dfrac{9}{7}$ and $\hat{\theta}' = \dfrac{4}{7} \Rightarrow \hat{\theta} = \begin{bmatrix} 4/7 \\ 9/7 \end{bmatrix}$

Best line (in the "least squares" sense) through the given data is

$$\boxed{\frac{4}{7}x + \frac{9}{7}}$$

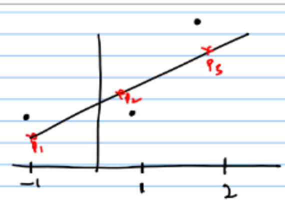
Projections: $\quad p_1 = \frac{4}{7}(-1) + \frac{9}{7} = \frac{5}{7}, \quad p_2 = \frac{13}{7}, \quad p_3 = \frac{17}{7}$

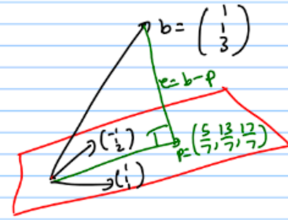
The original data is not on a line, so $E^2 > 0$

$$E^2 = \|b - A\hat{\theta}\|^2 = \|e\|^2$$

$$e = \left[1 - \left(-\frac{4}{7} + \frac{9}{7}\right)\right], \left[1 - \left(\frac{4}{7} + \frac{9}{7}\right)\right], \left[3 - \left(\frac{8}{7} + \frac{9}{7}\right)\right]$$

$$= \left(+\frac{2}{7}, -\frac{6}{7}, \frac{4}{7}\right)$$





$e = \left(+\frac{2}{7}, \frac{-6}{7}, \frac{4}{7}\right)$

$e \perp (-1, 1, 2)$ and $e \perp (1, 1, 1)$

QN : 1,3

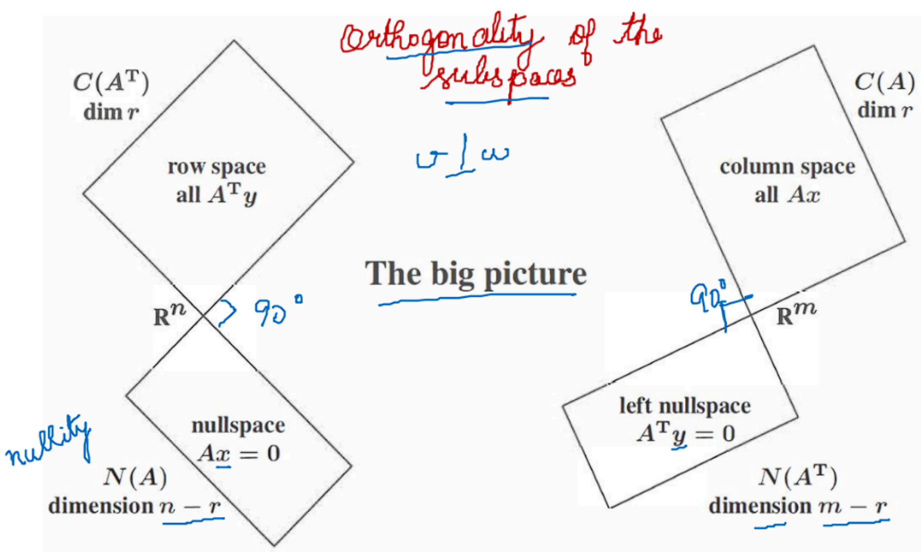
Tutorial 3.1 : Four fundamental vector subspaces

Four fundamental subspaces

Suppose **A** is a $m \times n$ matrix.

$$\begin{array}{cc} & n \\ A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} & m \end{array}$$

1. The column space is C(**A**), a subspace of $\mathbb{R}^m$.
2. The row space is C($\mathbf{A}^T$), a subspace of $\mathbb{R}^n$.
3. The nullspace is N(**A**), a subspace of $\mathbb{R}^n$. $\boxed{Ax=0}$
4. The left nullspace is N($\mathbf{A}^T$), a subspace of $\mathbb{R}^m$. $\boxed{A^T y=0}$



Dimension ?

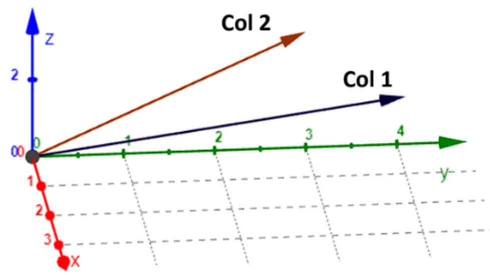
Rank ?

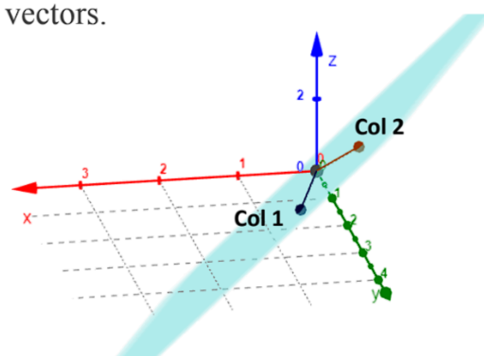
Rank nullity

$\boxed{\text{rank} + \text{nullity} = n}$

Column space visualization

- Consider a matrix: $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$
- C($\mathbf{A}$): Linear combinations of the column vectors.





https://www.geogebra.org/m/xk4qpm7c

Example 1

Obtain the four fundamental spaces of **A** and find its rank and nullity.

$$A = \begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 2 & 0 & 2 & 4 & -2 \\ 2 & 2 & 4 & 0 & 2 \\ 4 & 4 & 8 & 12 & 8 \end{bmatrix}$$

Column space

pivot

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 2 & 0 & 2 & 4 & -2 \\ 2 & 2 & 4 & 0 & 2 \\ 4 & 4 & 8 & 12 & 8 \end{bmatrix}$$

Gaussian elimination

Row echelon form

$R_2 - R_1$
$R_3 - R_1$
$R_4 - 2R_1$

$$\begin{pmatrix} 2 & 4 & 6 & 8 & 10 \\ 0 & -4 & -4 & -4 & -12 \\ 0 & -2 & -2 & -8 & -8 \\ 0 & -4 & -4 & -4 & -12 \end{pmatrix}$$

$R_3 - \frac{1}{2}R_2$
$R_4 - R_2$

column space

$$C(A) = \text{span}\left\{\begin{bmatrix}2\\2\\2\\4\end{bmatrix}, \begin{bmatrix}4\\0\\2\\4\end{bmatrix}, \begin{bmatrix}8\\4\\0\\12\end{bmatrix}\right\}$$

pivots

$$\begin{pmatrix} \boxed{2} & 4 & 6 & 8 & 10 \\ 0 & \boxed{-4} & -4 & -4 & -12 \\ 0 & 0 & 0 & \boxed{-6} & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon form

$\boxed{\text{rank} = 3} \Rightarrow \dim C(A)$

Null space

$N(x) = \{x \mid Ax = 0\}$ → $Ux = 0$

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 0 & -4 & -4 & -4 & -12 \\ 0 & 0 & 0 & -6 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

The homogeneous system of linear equations are:

$2x_1 + 4x_2 + 6x_3 + 8x_4 + 10x_5 = 0$ — (1)

$-4x_2 - 4x_3 - 4x_4 - 12x_5 = 0$ — (2)

$-6x_4 - 2x_5 = 0$ — (3)

Set $x_3 = 1$; $x_5 = 0$

$x_4 = 0$

Sub x_3, x_4 & x_5 in (2)

$-4x_2 - 4 = 0$

$x_2 = -1$

Sub x_2, x_3, x_4, x_5 in (1)

$2x_1 + 4(-1) + 6 = 0$

$x_1 = -1$

$$u = \begin{bmatrix}-1\\-1\\1\\0\\0\end{bmatrix}$$

set $x_3 = 0$; $x_5 = 1$

$$v = \begin{bmatrix}5/3\\-8/3\\0\\-1/3\\1\end{bmatrix}$$

$N(A) = \text{span}\{u, v\}$

$\dim N(A) = 2 \Rightarrow$ nullity

Row space $\rightarrow C(A^T)$

pivot

$$A^T = \begin{bmatrix} 2 & 2 & 2 & 4 \\ 4 & 0 & 2 & 4 \\ 6 & 2 & 4 & 8 \\ 8 & 4 & 0 & 12 \\ 10 & -2 & 2 & 8 \end{bmatrix}$$

$R_2 - 2R_1$
$R_3 - 3R_1$
$R_4 - 4R_1$
$R_5 - 5R_1$

$$\begin{pmatrix} 2 & 2 & 2 & 4 \\ 0 & -4 & -2 & -4 \\ 0 & -4 & -2 & -4 \\ 0 & -4 & -8 & -4 \\ 0 & -12 & -8 & -12 \end{pmatrix} \xrightarrow[R_5 - 3R_2]{R_3 - R_2,\ R_4 - R_2} \begin{pmatrix} 2 & 2 & 2 & 4 \\ 0 & -4 & -2 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & -2 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_5} \begin{pmatrix} 2 & 2 & 2 & 4 \\ 0 & -4 & -2 & -4 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$R_4 - 3R_3$

pivots

$$\begin{pmatrix} 2 & 2 & 2 & 4 \\ 0 & -4 & -2 & -4 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$R(A)$
$\downarrow$

$$\text{span}\left\{ \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \\ 10 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \\ 0 \\ 2 \end{bmatrix} \right\}$$

$\dim R(A) = 3$
$\downarrow$
r

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Left null space

$N(A^T) = \{ y \mid A^T y = 0 \}$

$$\begin{bmatrix} 2 & 2 & 2 & 4 \\ 0 & -4 & -2 & -4 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The system of linear equations are:

$2y_1 + 2y_2 + 2y_3 + 4y_4 = 0$ — (1)

$-4y_2 - 2y_3 - 4y_4 = 0$ — (2)

$-2y_3 = 0$

$y_3 = 0$

Set $y_4 = 1$

$-4y_2 - 4 = 0$

$y_2 = -1$

$2y_1 + 2(-1) + 4(1) = 0$

$y_1 = -1$

$$y = \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ Left nullspace of A
$\downarrow$

$$\text{Span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Tutorial 3.2 : Row space computation using reduced row echelon form and Solution to Ax = b

Example 2

For the matrix **B** find its row space by computing its reduced row echelon form.

$$B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ -2 & 1 & 2 \end{bmatrix}$$

Solution

$$B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ -2 & 1 & 2 \end{bmatrix} \xrightarrow[R_3+2R_2]{R_2-2R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 3 & 6 \end{pmatrix} \xrightarrow{R_3-3R_2} \underbrace{\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}}_{\text{Row echelon form}} \xrightarrow{R_1-R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{Reduced row echelon form}$$

Row space of B

$$\text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Example 3

Consider the system of equations given below:

$$x_1 + 2x_2 - 2x_3 = b_1$$
$$2x_1 + 5x_2 - 4x_3 = b_2$$
$$4x_1 + 9x_2 - 8x_3 = b_3$$

i. Under what condition on b_1, b_2 and b_3, is the system solvable?

ii. Check whether $b_1 = 3$, $b_2 = -1$ and $b_3 = 5$ satisfies the condition or not?

Solution

$$x_1 + 2x_2 - 2x_3 = b_1$$
$$2x_1 + 5x_2 - 4x_3 = b_2$$
$$4x_1 + 9x_2 - 8x_3 = b_3$$

coefficient matrix

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 4 & 9 & -8 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Augmented matrix $= [Ab]$

$$\left(\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 2 & 5 & -4 & b_2 \\ 4 & 9 & -8 & b_3 \end{array}\right) \xrightarrow[R_3 - 4R_1]{R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 1 & 0 & b_3 - 4b_1 \end{array}\right) \xrightarrow{R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & -2 & b_1 \\ 0 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & \underline{b_3 - b_2 - 4b_1} \end{array}\right)$$

condition : $b_3 - b_2 - 4b_1 = 5 - (-1) - 4(2) = 6 - 6 = 0$ ✓

$b_1 = 3$, $b_2 = -1$ and $b_3 = 5$

$$Ax = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix} \quad b \in C(A)$$

Tutorial 3.3 : Orthogonality, projection and least squares method

Example 1

Let $S = \{(1, 2, 4, 0)^T, (-2, 3, -1, 0)^T, (0, 2, 6, -1)^T\}$. Which pair(s) of vectors in this given set are orthogonal?

Solution

$u \cdot v = -2 + 6 - 4$
$= 0$ ✓

$v \cdot w = 6 - 6$
$= 0$ ✓

$S = \{\underset{u}{(1, 2, 4, 0)^T}, \underset{v}{(-2, 3, -1, 0)^T}, \underset{w}{(0, 2, 6, -1)^T}\}$

$u \cdot w = 4 + 24$
$= 28$ ×

$\therefore$ (u, v) and (v, w) are orthogonal pairs.

Example 2

i. Find the projection matrix for $a = \begin{bmatrix} 2 \\ -1 \\ 2 \\ 3 \end{bmatrix}$.

ii. Obtain the projection of $b = \begin{bmatrix} 1 \\ 3 \\ -2 \\ 5 \end{bmatrix}$ onto a and compute the error.

Solution

i) Projection matrix of $a = \begin{pmatrix} 2 \\ -1 \\ 2 \\ 3 \end{pmatrix}$ $\qquad \mathbb{P} = \dfrac{aa^T}{a^Ta}$

$$aa^T = \begin{pmatrix} 2 \\ -1 \\ 2 \\ 3 \end{pmatrix} \cdot (2\ -1\ 2\ 3) = \begin{pmatrix} 4 & -2 & 4 & 6 \\ -2 & 1 & -2 & -3 \\ 4 & -2 & 4 & 6 \\ 6 & -3 & 6 & 9 \end{pmatrix} \qquad a^Ta = (2\ -1\ 2\ 3)\begin{pmatrix} 2 \\ -1 \\ 2 \\ 3 \end{pmatrix} = 4+1+4+9 = 18$$

$$\mathbb{P} = \begin{pmatrix} 2/9 & -1/9 & 2/9 & 1/3 \\ -1/9 & 1/18 & -1/9 & -1/6 \\ 2/9 & -1/9 & 2/9 & 1/3 \\ 1/3 & -1/6 & 1/3 & 1/2 \end{pmatrix}$$

ii) Projection of $b = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 5 \end{pmatrix}$ onto a

$$p = \mathbb{P}.b = \begin{pmatrix} 10/9 \\ -5/9 \\ 10/9 \\ 5/3 \end{pmatrix}$$

compute error, $e = b - p$

$$e = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 5 \end{pmatrix} - \begin{pmatrix} 10/9 \\ -5/9 \\ 10/9 \\ 5/3 \end{pmatrix} = 1/9 \begin{pmatrix} -1 \\ 32 \\ 28 \\ 30 \end{pmatrix}$$

Least squares method

- To determine the 'line of best fit' for a set of data.
- How?
- By minimizing the sum of the offsets or residuals of points from the plotted line.
- Represents general trend of the data.
- Used for regression analysis.

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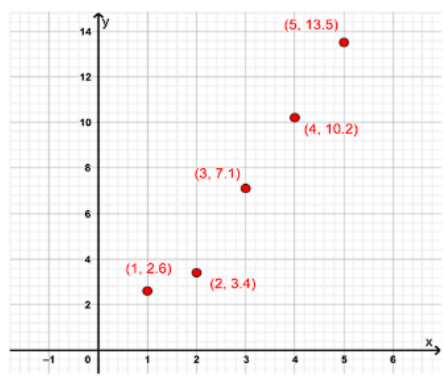
Example 3

Build a model that studies the relationship between x and y given in the Table below using least squares method.

x	1	2	3	4	5
y	2.6	3.4	7.1	10.2	13.5



Solution

- Least Squares method: $\boldsymbol{A}^T\boldsymbol{A}\hat{\boldsymbol{x}} = \boldsymbol{A}^T\mathrm{b}$ ✓

$$\hat{x} = \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} \begin{array}{l} \rightarrow \text{slope} \\ \rightarrow \text{intercept} \end{array}$$

$$A = \begin{array}{c} \begin{array}{cc} x\downarrow & \end{array} \\ \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{bmatrix} \end{array} \quad b = \begin{array}{c} y\downarrow \\ \begin{bmatrix} 2.6 \\ 3.4 \\ 7.1 \\ 10.2 \\ 13.5 \end{bmatrix} \end{array}$$

Find A^TA and A^Tb

$$A^T = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{bmatrix} \qquad A^Tb = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2.6 \\ 3.4 \\ 7.1 \\ 10.2 \\ 13.5 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 55 & 15 \\ 15 & 5 \end{bmatrix} \qquad A^Tb = \begin{bmatrix} 139 \\ 36.8 \end{bmatrix}$$

$$\begin{bmatrix} 55 & 15 \\ 15 & 5 \end{bmatrix} \begin{bmatrix} \hat{\theta}' \\ \hat{\theta}'' \end{bmatrix} = \begin{bmatrix} 139 \\ 36.8 \end{bmatrix}$$

$$55\hat{\theta}' + 15\hat{\theta}'' = 139$$

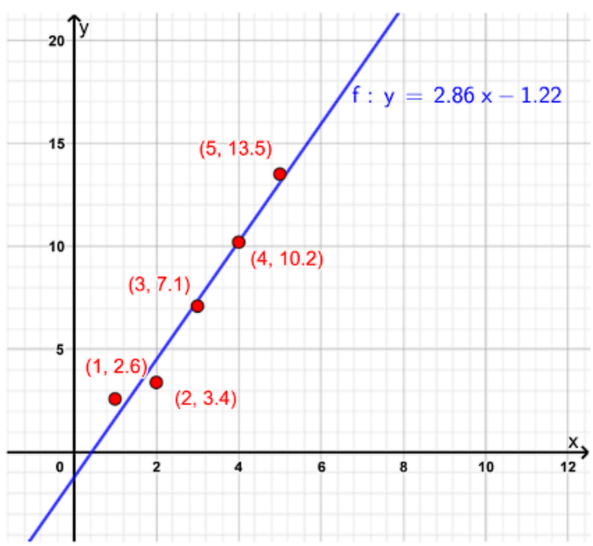
$$15\hat{\theta}' + 5\hat{\theta}'' = 36.8$$

Solving these equations;

$$\hat{\theta}' = 2.86$$

$$\hat{\theta}'' = -1.22$$

Best line through the given data is : $y = 2.86\,x - 1.22$



Week 3 : Live session 1

1. Vector spaces

A vector space is a set V along with two operations, $+$ and $\cdot$, addition and scalar multiplication that obeys certain axioms:

Addition, $+$

- Commutativity: $x + y = y + x$
- Associativity: $x + (y + z) = (x + y) + z$
- Additive identity $0 \in V,\ \ x + 0 = x$
- Additive inverse: $x + (-x) = 0$

Scalar multiplication, $\cdot$

- Multiplicative identity, $1 \cdot x = x$
- Associativity, $a \cdot (b \cdot x) = (ab) \cdot x$
- Distributivity:
  - $(a + b) \cdot x = a \cdot x + b \cdot x$
  - $a \cdot (x + y) = a \cdot x + a \cdot y$

Examples of vector spaces:

- $\mathbb{R}^n$: the set of all $n-$tuples
  - $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \ \cdots \ \times \mathbb{R}$
  - $(x_1, \ \cdots, x_n) \in \mathbb{R}^n$, where $x_i \in \mathbb{R}$
  - $\mathbb{R}^2$, the plane
  - $\mathbb{R}^3$, the space
- $\mathbb{C}^n$: the set of all $n-$tuples
  - $\mathbb{C}^n = \mathbb{C} \times \mathbb{C} \times \ \cdots \ \times \mathbb{C}$
  - $\mathbb{C}$ is the set of all complex numbers
  - $(x_1, \ \cdots, x_n)$, where $x_i \in \mathbb{C}$

- $\mathbb{C}^2$
    * $(2+3i, 5i)$
        - $i^2 = -1$ or $i = \sqrt{-1}$

2. Subspaces

Let us consider $\mathbb{R}^2$. Is $\{(0,1),(1,0),(0,0)\}$ a subspace of $\mathbb{R}^2$?

No. It is not.

- First, a subspace should be a subset of a vector space.
- Secondly, a subspace should be a vector space in its own right.

How do you verify if a subset S is a subspace of a $\mathbb{R}^2$?
- $(0,0) \in S$
- S should be closed under addition and scalar multiplication

What does closure mean?
- If you add two vectors, the result should also be in the set.
- If you multiply a vector by a scalar, the result should be in the set.

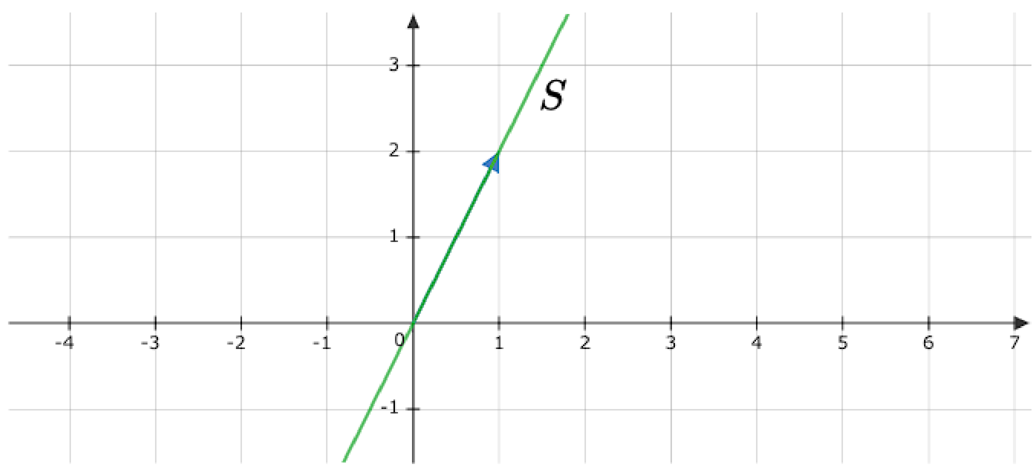
$$S = \{(0,1),(1,0),(0,0)\}$$

S is not a subspace because $(0,1)+(1,0)=(1,1) \notin S$

S is not a subspace because $(0,1)+(1,0)=(1,1) \notin S$

Let's look at some subspaces of $\mathbb{R}^2$

- all lines passing through the origin are subspaces of $\mathbb{R}^2$
- $S=\{\alpha \cdot (1,2) : \alpha \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$



Verifying if S is closed under addition:

$$\alpha \cdot (1,2)+\beta \cdot (1,2)=(\alpha+\beta)\cdot(1,2)=\gamma \cdot (1,2)$$

Every vector space has two trivial subspaces. For example, $\mathbb{R}^2$:

- $\{(0,0)\}$
- $\mathbb{R}^2$

3. Linear independence and span

3.1. Span

Span of S is the set of all possible linear combinations of vectors in S. For example:

$$S = \{(1,1),(2,3)\}$$

The following is one linear combination:

$$2 \cdot (1,1) - 5(2,3)$$

The span is the set of all possible linear combinations:

$$\text{span}(S) = \{\alpha \cdot (1,1) + \beta \cdot (2,3) : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$$

In general, if

$$S = \{v_1, \cdots, v_n\}$$

$$\text{span}(S) = \{\alpha_1 \cdot v_1 + \cdots + \alpha_n v_n : \alpha_i \in \mathbb{R}\}$$

$\mathbb{R}$ or $\mathbb{C}$ are fields

Properties of span:

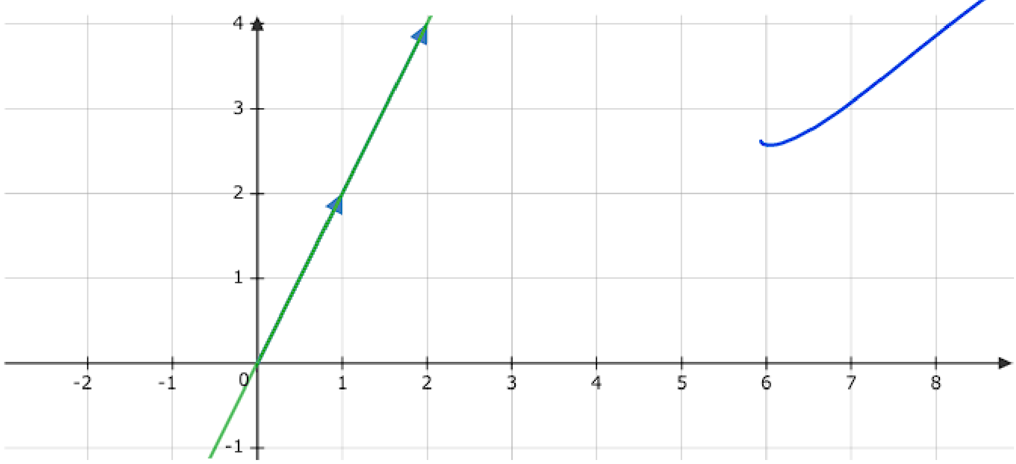
- span(S) is a subspace
- span(S) is the smallest subspace containing S
- span$(\{\ \}) = \{0\}$, the trivial subspace

3.2. Linear Independence

Let's look at the following vectors:

$$u = (1, 1)$$
$$v = (2, 2)$$

We can write $v = 2u$. If you have two vectors $u, v \in \mathbb{R}^2$, they are called linearly dependent if one vector can be written as a multiple of the other. $v = \alpha \cdot u$ or $u = \alpha \cdot v$ for some $\alpha \in \mathbb{R}$, then $\{u, v\}$ is linearly dependent.



Now consider these vectors in $\mathbb{R}^3$:

$$\begin{aligned} u &= (1,0) \\ v &= (0,1) \\ w &= (1,1) \end{aligned}$$

Notice $w = u + v$. We say that $\{u, v, w\}$ is linearly dependent.

We say that a set of vectors $S = \{v_1, \cdots, v_n\}$ is linearly dependent if there is some vector $v_i \in S$ that is in the span of the remaining vectors.

$$v_i = \alpha_1 v_1 + \cdots + \alpha_{i-1} \cdot v_{i-1} + \alpha_{i+1} v_{i+1} + \cdots + \alpha_n v_n$$

There is alternative way for defining dependence/independence. First you take any linear combination and set it to zero:

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$$

Is there some choice of α_i that will give me zero? This will always happen when $\alpha_1 = \cdots = \alpha_n = 0$. Linear independence demands that there is no linear combination other than the trivial one which will result in the zero vector.

$$
\begin{aligned}
u &= (1,0) \\
v &= (0,1) \\
w &= (1,1)
\end{aligned}
$$

$$\alpha_1(1,0) + \alpha_2(0,1) + \alpha_3(1,1) = (0,0)$$

- $\alpha_1 = \alpha_2 = \alpha_3 = 0$
- $\alpha_1 = 1, \alpha_2 = 1, \alpha_3 = -1 \rightarrow$ non-trivial combination

The moment you have a non-trivial combination, the set of vectors becomes dependent.

A set $S = \{v_1, \cdots, v_n\}$ is **linearly independent** if

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0 \implies \alpha_i = 0,\ 1 \leqslant i \leqslant n$$

If $0 \in S$ then, S is linearly dependent.

3.3. Spanning set

A spanning set S of a vector space V is a set whose span is the entire vector space, meaning, span$(S) = V$. S is said to span V. For example, $\{(1,0),(0,1)\}$ spans $\mathbb{R}^2$, $\{(1,0,0),(0,1,0),(0,0,1),(1,2,3)\}$ spans $\mathbb{R}^3$.

4. Basis and dimension

A basis of a vector space V is a linearly independent spanning set. Therefore, if B is a basis for V:

- B has to be linearly independent
- span$(S) = V$

Examples:

- $B = \{(1,0),(0,1)\}$ is a basis for $\mathbb{R}^2$
- $B = \{(1,0,\cdots,0),\cdots,(0,0,\cdots.,1)\}$ is a basis for $\mathbb{R}^n$
- The above bases are examples of standard bases

Properties/Observations

- Basis is not unique
  - $\{(1,2),(2,1)\}$ is another basis for $\mathbb{R}^2$
- Minimally spanning set: If you remove a vector from a basis, it is no longer spanning set
- Maximally linearly independent set: If you add a vector to a basis, it is no longer linearly independent
- Given any linearly independent set, you can extend it to form a basis:
  - $B' = \{(1,0,0),(0,0,1)\}$, $B = B \cup \{(0,1,0)\}$
- Given any spanning set, you can shrink it to form a basis:

  - $B' = \{(1,0),(0,1),(2,3),(5,6)\}$
  - Throw any two vectors out to get a basis
- 0 can never be an element of a basis
- All bases for a vector space are of the same size

The dimension of a vector space is the number of vectors in the basis. That is, $\dim(V) = |B|$. In this course we will study what are called finite dimensional vector spaces.

Does anyone know an infinite dimensional vector space? The set of all polynomials is an example.

$$a_0 + a_1 x + \cdots + a_n x^n \in \mathcal{P}$$

$$B = \left\{1, x, x^2, \cdots\right\}$$

Convention, the basis for the trivial subspace $\{0\}$ is the empty set $\{\ \}$. Therefore, $\dim\{0\} = 0$. $\{\ \}$ is a linearly independent spanning set for $\{0\}$ and therefore a basis.

An example of a one-dimensional vector space would be any line passing through the origin in $\mathbb{R}^2$.

5. Dot product

So far what all things can we do with vectors?

- add
- scale them by scalars

Why do we need dot products? To do geometry.

- compute lengths of vectors
- compute angles between vectors

$$\langle x, y \rangle = x^T y$$

$$x = (x_1, x_2), y = (y_1, y_2)$$

$$\boxed{x^T y = x_1 y_1 + x_2 y_2}$$

Properties:

- non-negativity:
  - $x^T x \geqslant 0$
  - $x^T x = 0$ if and only if $x = 0$
  - $x_1^2 + x^2 \geqslant 0$
- linearity
  - $(x + y)^T z = x^T z + y^T z$
    $(\alpha \cdot x)^T y = \alpha \cdot (x^T y)$
- symmetry
  - $x^T y = y^T x$

Note: It is actually "conjugate symmetry" for complex vector spaces. In complex vector spaces, simple symmetry doesn't work for the complex dot product.

6. Norm and angle

The norm of a vector is the length of the vector. We have what is called the dot-product induced norm or the norm induced by the dot product.

$$||x|| = \sqrt{x^T x}$$

The angle between two vectors x, y in $\mathbb{R}^2$ is given by:

$$\theta = \cos^{-1}\left(\frac{x^T y}{||x|| \cdot ||y||}\right)$$

$$\cos\theta = \frac{x^T y}{||x|| \cdot ||y||}$$

The above relation is often called the cosine similarity in the ML literature. It measures the similarity between two vectors. Vectors are features or attributes representing data. For example, let's say you are describing a house:

$$x = \begin{bmatrix} \text{area} \\ \text{number of rooms} \end{bmatrix} = \begin{bmatrix} 1000 \\ 3 \end{bmatrix}$$

$$y = \begin{bmatrix} \text{area} \\ \text{number of rooms} \end{bmatrix} = \begin{bmatrix} 1200 \\ 2 \end{bmatrix}$$

Cosine similarity measures the similarity between two houses:

$$\frac{x^T y}{||x|| \cdot ||y||}$$

What is the range of values of the above expression? $[-1, 1]$. In general, the higher the cosine-similarity, more similar the vectors or the data-points (houses) you are comparing. If $x = y$, the cosine-similarity will be 1.

Cauchy-Schwartz inequality:

$$x^T y \leqslant ||x|| \cdot ||y||$$

How do you convert $(1, 2, 3)$ into a unit-norm vector?

- First find the norm.
- Divide the vector by the norm.

$$||(1, 2, 3)|| = \sqrt{1^2 + 2^2 + 3^2}$$

$$\frac{(1, 2, 3)}{\sqrt{14}}$$

7. Orthogonality

Two vectors u, v are orthogonal (perpendicular) if $u^T v = 0$. For example, the following pairs are orthogonal:

- $(1, 0) \perp (0, 1) \rightarrow$ perp
- $(1, 2, 1) \perp (-1, 1, -1)$
- $(0, 0, 0) \perp u$ for all $u \in \mathbb{R}^3$. In general, $0 \perp V$

Cosine similarity between two orthogonal vectors is 0. 0 is the only vector that is orthogonal to itself.

8. Orthogonality and linear independence

A set of vectors $S = \{v_1, \cdots, v_n\}$ is said to be pair-wise orthogonal if $v_i^T v_j = 0$ for $i \neq j$. If S is a collection of non-zero orthogonal vectors, then S is linearly independent.

Take any linear combination and set it to zero. To show:

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0 \Longrightarrow \alpha_i = 0$$

$$v_i^T(\alpha_1 v_1 + \cdots + \alpha_n v_n) = v_i^T 0$$

$$\alpha_1 v_i^T v_1 + \cdots + \alpha_{i-1} v_i^T v_{i-1} + \alpha_i v_i^T v_i + \cdots + \alpha_n v_i^T v_n = 0$$

$$\alpha_i ||v_i||^2 = 0 \Longrightarrow \alpha_i = 0$$

Therefore, we have $\alpha_1 = \cdots = \alpha_n = 0$ and that shows that S is linearly independent.

Orthogonal sets are linearly independent provided the set doesn't have the zero vector.

9. Orthonormal basis

$S = \{v_1, \cdots, v_n\}$ is an orthonormal basis for $\mathbb{R}^n$ if:

$$v_i^T v_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

The vectors in an orthonormal basis are pair-wise orthogonal and have unit norm. For example:

- $\{(1,0),(0,1)\}$ is an orthonormal basis for $\mathbb{R}^2$
- $\left\{\dfrac{(3,4)}{5}, \dfrac{(-4,3)}{5}\right\}$ is an orthonormal basis for $\mathbb{R}^2$

Note that $\{(3,4),(-4,3)\}$ is an orthogonal basis but not orthonormal.

Now, any vector in $\mathbb{R}^n$ can be expressed as:

$$\boxed{v = \left(v^T v_1\right) v_1 + \cdots + \left(v^T v_n\right) v_n}$$

For example, consider $B = \left\{\frac{(3,4)}{5}, \frac{(-4,3)}{5}\right\}$. How do you represent $(1,2)$ in this basis?

$$v = (1,2), \quad \begin{aligned} v_1 &= \frac{(3,4)}{5} \\ v_2 &= \frac{(-4,3)}{5} \end{aligned}$$

$$v = \alpha_1 v_1 + \alpha_2 v_2$$

$$\begin{aligned} \alpha_1 &= v^T v_1 & \alpha_2 &= v^T v_2 \\ &= \frac{1}{5} \cdot \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}, & &= \frac{1}{5} \cdot \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \end{bmatrix} \\ &= \frac{11}{5} & &= \frac{2}{5} \end{aligned}$$

$$(1,2) = \frac{11}{5} \cdot \left[\frac{(3,4)}{5}\right] + \frac{2}{5} \cdot \left[\frac{(-4,3)}{5}\right]$$

Verify:

$$\frac{(33-8, 44+6)}{25} = \frac{(25,50)}{25} = (1,2)$$

10. Four Fundamental Subspaces

A matrix A of shape $m \times n$ has four subspaces associated with it:

- Column space
- Row space
- Nullspace
- Left nullspace

10.1. Column space

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$

The column space is the span of the columns:

$$\mathcal{C}(A) = \text{span}\{a_1, \cdots, a_n\}$$

- The column space is a subspace of $\mathbb{R}^m$
- The dimension of the column space is called column rank

10.2. Row space

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

$$\mathcal{R}(A) = \text{span}\left\{a_1^T, \cdots, a_m^T\right\}$$

- The row space is a subspace of $\mathbb{R}^n$
- The dimension of the row space is called the row rank
- The row rank is equal to the column rank. We will just use the term "rank" of the matrix.
- We have $\mathcal{C}\left(A^T\right) = \mathcal{R}(A)$ because the operation of transpose swaps the rows and columns. What was a row before now becomes a column and vice-versa.

Rank of a matrix

- The number of linearly independent columns.
- The number of linearly independent rows.

10.3. Null space

The null space of a matrix A is the set of all solutions to the homogeneous system $Ax = 0$.

$$\mathcal{N}(A) = \left\{x : Ax = 0, x \in \mathbb{R}^n\right\}$$

The dimension of the null space is called nullity. It is known that:

$$\boxed{\text{rank} + \text{nullity} = \text{number of columns}}$$

This is the rank-nullity theorem.

10.4. Left null space

The set of all solutions to the equation $y^T A = 0$ is the left null space:

$$\mathcal{N}(A^T) = \left\{y : A^T y = 0, y \in \mathbb{R}^m\right\}$$

$$y^T A = 0 \implies \left(y^T A\right)^T = 0 \implies A^T y = 0$$

Remember the following pairing:

- Null space and the row space are subspaces of $\mathbb{R}^n$
- Column space and the left null space are subspaces of $\mathbb{R}^m$

10.5. Algorithm to compute the four subspaces

We will do this with the help of an example:

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

Row space

- Reduce the matrix to its row echelon form using elementary row operations:
  - Scale a row by a **non-zero** constant
  - Swap rows
  - Add a constant times one row to **another** row

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- The first non-zero element in each row is the pivot.
- The non-zero rows of the row-echelon matrix is a basis for the row space.
- Every row operation doesn't disturb the row space. The row space is preserved across all row operations.

Row space is given by span$\{(1,0,1,2),(0,1,3,1)\}$

We are trying to show the non-zero rows in the row-echelon form are linearly independent and span the row space.

$$\begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \end{bmatrix} \rightarrow$$

$$\alpha(1,0,1,2)+\beta(0,1,3,1)=(0,0,0,0)$$

$$(\alpha, \beta, \alpha+3\beta, 2\alpha+\beta)=(0,0,0,0)$$

$$\implies \alpha=\beta=0$$

Why is this case? The pivot in every row is to the right of the pivot in the previous row.

Column space

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Identify the pivot columns: the columns that contain the pivots
- Go back to the original matrix and extract the columns corresponding to the pivot column
- These columns form a basis for the column space

$$\mathcal{C}(A) = \text{span}\{(1,0,1),(0,1,1)\}$$

Verifying that we have a spanning set for the column space.

$$(1,3,4) = 1(1,0,1) + 3(0,1,1)$$

$$(2,1,3) = 2(1,0,1) + 1(0,1,1)$$

So far we have found out the rank is 2.

Null space

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Homogeneous system:

We want to find a basis for the nullspace.

$$\begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \color{red}{x_1} \\ \color{red}{x_2} \\ \color{green}{x_3} \\ \color{green}{x_4} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- x_1, x_2 are the pivot variables
- x_3, x_4 are the free variables

Set $x_3 = 1, x_4 = 0$

$$x_2 = -3, x_1 = -1$$

Therefore, one vector in the basis is $(-1, -3, 1, 0)$.

Set $x_3 = 0, x_4 = 1$

$$x_2 = -1, x_1 = -2$$

Therefore, the other vector in the basis is $(-2, -1, 0, 1)$.

Finally, a basis for the nullspace of A is:

$$\{(-1, -3, 1, 0), (-2, -1, 0, 1)\}$$

and

$$\mathcal{N}(A) = \text{span}\{(-1, -3, 1, 0), (-2, -1, 0, 1)\}$$

Verify if these two are actually in the nullspace.

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- The row space is orthogonal to the null space.
- The column space is orthogonal to the left null space.

Week 3 : Live session 2

1. System of Linear Equations

There are m linear equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + \ \cdots\ + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \ \cdots\ + a_{mn}x_n &= b_m \end{aligned}$$

Now if you collect the coefficients in a matrix and the RHS and the unknowns in vectors:

$$Ax = b$$

- A is of shape $m \times n$
- $x \in \mathbb{R}^n$
- $b \in \mathbb{R}^m$

One can look at this system in two ways. After professor Gilbert Strang, we have the row picture and the column picture.

1.1. Row picture

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

You can treat each row of A as a row vector. The i^{th} row is $a_i^T \in \mathbb{R}^n$.

$$\begin{aligned} a_1^T x &= b_1 \\ &\vdots \\ a_m^T x &= b_m \end{aligned}$$

1.2. Column picture

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$

When A is expressed as a collection of n column-vectors, then Ax is the linear combination of the columns of A. Therefore, the LHS is a linear combination of columns.

$$x_1 a_1 + \cdots + x_n a_n = b$$

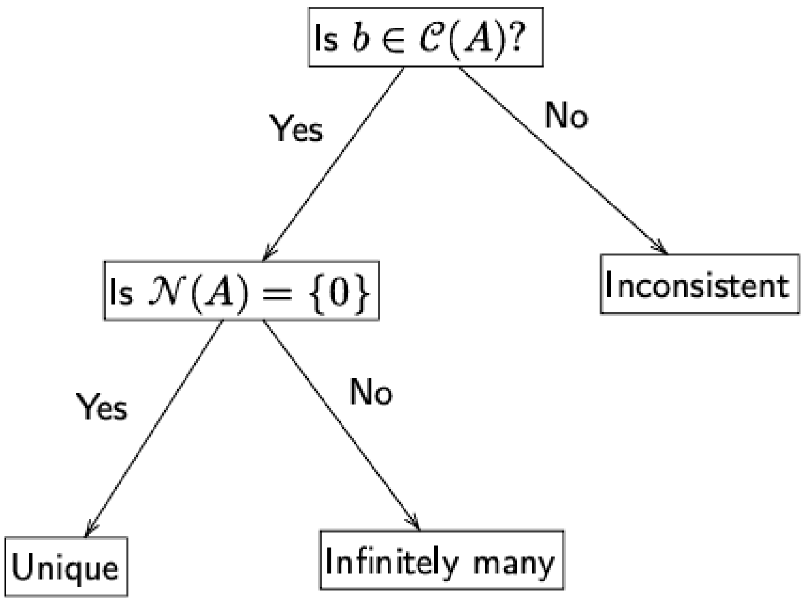
If $Ax = b$ should have a solution, if it is to be consistent, that is, then b should be in the column space of A. Given its importance, we make it more prominent:

> **Necessary and sufficient condition for consistency**: A system of linear equations, $Ax = b$, is consistent if and only if b is in $\mathcal{C}(A)$, where $\mathcal{C}(A)$ is the column space of A.

Recall that a system of linear equations is consistent if it has at least one solution. The following is an example of a system that is **not consistent**.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

We can come up with the following tree. Refer to the remainder of the live session to understand how this tree is constructed.



2. $Ax = 0$, homogeneous system

Quickly, the algorithm:

- Find the RREF of A
- Identify pivots and pivot columns
- Identify free variables and pivot variables
- Follow the usual procedure; refer to this document for the full details

The set of all solutions to $Ax = 0$ is called the nullspace of A and denoted by $\mathcal{N}(A)$.

- $\mathcal{N}(A)$ is subspace of $\mathbb{R}^n$
- If $Ax = 0$ has a unique solution, then $\mathcal{N}(A) = \{0\}$
- The following are equivalent to the statement that $\mathcal{N}(A) = \{0\}$
  - The columns are linearly independent.
  - There are no free (independent) variables while solving $Ax = 0$
  - There are n pivot variables when solving $Ax = 0$
  - The matrix has full column rank

  - Warning: If in addition the matrix is A square, then this will happen when $|A| \neq 0$ or when A is invertible.

Error Alert:

The following erroneous statements were made in the live session.

If the columns span $\mathbb{R}^n$ then $\mathcal{N}(A) = \{0\}$.
If the columns form a basis for $\mathbb{R}^n$ then $\mathcal{N}(A) = \{0\}$

- The first error is that the columns are in $\mathbb{R}^m$ and not $\mathbb{R}^n$.
- Secondly, it is sufficient for the columns to be linearly independent. It is not necessary that they span $\mathbb{R}^m$ for the nullspace to be $\{0\}$ or that they are a basis for $\mathbb{R}^m$.
- These statements are true only if $m = n$.

3. $Ax = b$

The goal is to find the set of all solutions to $Ax = b$ assuming that it is consistent.

Check later:

$$(1, 2, 0, 1)$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

Step-1: Augmented matrix

$$\begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 2 & 1 & 1 & 1 & & 5 \\ 1 & 2 & 1 & 1 & & 6 \end{bmatrix}$$

Step-2: Row reduction

Also called forward elimination.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 2 & 1 & 1 & 1 & & 5 \\ 1 & 2 & 1 & 1 & & 6 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 1 & 2 & 1 & 1 & & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 1 & 2 & 1 & 1 & & 6 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 0 & 2 & -1 & 1 & & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 0 & 2 & -1 & 1 & & 5 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 0 & 0 & 5 & -1 & & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 0 & 0 & 5 & -1 & & -1 \end{bmatrix} \xrightarrow{R_3 \to \frac{R_3}{5}} \begin{bmatrix} 1 & 0 & 2 & 0 & & 1 \\ 0 & 1 & -3 & 1 & & 3 \\ 0 & 0 & 1 & -1/5 & & -1/5 \end{bmatrix}$$

Step-3: Identify the pivots and the pivot columns

$$\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & b \end{array} \\ \begin{bmatrix} \mathbf{1} & 0 & 2 & 0 & 1 \\ 0 & \mathbf{1} & -3 & 1 & 3 \\ 0 & 0 & \mathbf{1} & -1/5 & -1/5 \end{bmatrix} \end{array}$$

Step-4: Identify free and pivot variables

- Free variables: x_4
- Pivot variables: x_1, x_2, x_3

Step-5: Identify the nature of solutions

Since there is a free variable, the system has infinitely many solutions.

Step-6: Particular solution

The easiest way to find one solution is to set the free variables to zero and solve for the pivot variables using **backward substitution**.

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & & b \\ \mathbf{1} & 0 & 2 & 0 & & 1 \\ 0 & \mathbf{1} & -3 & 1 & & 3 \\ 0 & 0 & \mathbf{1} & -1/5 & & -1/5 \end{bmatrix}$$

$$\begin{aligned} x_1 &= 1 - 2x_3 \\ &= 1 + \frac{2}{5} \\ &= \frac{7}{5} \\ x_2 &= 3 + 3x_3 \\ &= 3 - \frac{3}{5} \\ &= \frac{12}{5} \\ x_3 &= -1/5 \\ x_4 &= 0 \end{aligned}$$

Therefore, a solution to $Ax = b$ is:

$$x_p = \left(\frac{7}{5}, \frac{12}{5}, \frac{-1}{5}, 0\right)$$

Error Alert:

I multiplied by 5 and called that a particular solution.

When solving $Ax = b$, one can't arbitrarily scale a solution. This makes sense only while solving $Ax = 0$, the homogeneous system.

x_p is called a particular solution to the system. Note that $Ax_p = b$.

What is the general solution?

If $x_n \in \mathcal{N}(A)$, then $Ax_n = 0$. What can you say about $x_p + x_n$? Is that a solution to the system $Ax = b$?

$$\begin{aligned} A(x_n + x_p) &= Ax_n + Ax_p \\ &= 0 + b \\ &= b \end{aligned}$$

Therefore, $x_n + x_p$ is a solution to $Ax = b$. One can show that:

Let $Ax = b$ be consistent and let x_p a particular solution. Then the set of all solution to $Ax = b$ is:

$$x_p + \mathcal{N}(A)$$

That is, any solution to $Ax = b$ can be written as a particular solution plus some vector from the nullspace of A.

If $Ax = b$ is consistent, when does $Ax = b$ have a unique solution?

- If the nullspace is $\{0\}$
- Refer to $Ax = 0$ discussion for other ways of representing this.

Remark: If A is invertible, then there is a unique solution and that is given as $x = A^{-1}b$.

4. $Ax \approx b$

$Ax = b$ is not always consistent, especially in a practical setting. In such cases, the best thing to do is to resort to an approximation.

Approximating scalar values, say $\sqrt{2}$

- 1.414
- 1.41
- 1.4

Goodness of approximation:

$$e = \sqrt{2} - 1.414$$

Smaller the error, better the approximation.

We now have an inconsistent system, therefore, we need to look for approximations:

$$\boxed{Ax \approx b}$$

Can I get close to b? Can I find an x that will get me as close to b as possible?

$$b = (1, 2, 3)$$

$$Ax' = (1.5, 2.5, 3.5)$$
$$Ax'' = (1.1, 2.1, 3.1)$$

For approximating vectors, what process are you going to follow?

We could use simple differences as a measure of the goodness of the approximation:

$$e' = (1.5 - 1) + (2.5 - 2) + (3.5 - 3)$$

But that has a problem as the differences might cancel each other out.

$$(1.5, 2, 2.5)$$

$$(1.5 - 1) + (2 - 2) + (2.5 - 3) = 0$$

Therefore, take the square of the differences:

$$\boxed{e = (1.5 - 1)^2 + (2.5 - 2)^2 + (3.5 - 3)^2}$$

The sum of squared errors (SSE).

5. Linear Regression and Least Squares

Consider the following problem:

$$\begin{bmatrix} Q_1 & Q_2 \\ 50 & 70 \\ 60 & 60 \\ 70 & 80 \\ 100 & 90 \end{bmatrix} \quad \begin{bmatrix} E \\ 80 \\ 60 \\ 60 \\ 80 \end{bmatrix}$$

Based on historical data of performance of previous term students in MLF, you as an admin want to predict how students in the current term will perform in the end-term based on their Q1, Q2 scores.

This is an example of a regression problem: you want to predict a real number.

- Features: Q1 and Q2 marks
- Label (target): end-term marks

If the model that you choose to represent the relationship between the labels and the features is linear, then you have linear regression.

$$\boxed{\widehat{E} = x_0 + x_1 Q_1 + x_2 Q_2}$$

$$\begin{aligned}
50x_1 + 70x_2 &\approx 80\\
60x_1 + 60x_2 &\approx 60\\
70x_1 + 80x_2 &\approx 60\\
100x_1 + 90x_2 &\approx 80
\end{aligned}$$

This is an overdetermined system. More equations than unknowns.

$$\begin{bmatrix} 50 & 70\\ 60 & 60\\ 70 & 80\\ 100 & 90 \end{bmatrix} \begin{bmatrix} x_1\\ x_2 \end{bmatrix} \approx \begin{bmatrix} 80\\ 60\\ 60\\ 80 \end{bmatrix}$$

$$\boxed{Ax \approx b}$$

- A holds the features of your data (comes from history)
  - The first column is Q1 scores
  - The second column is Q2 scores
- b holds the true labels (comes from history)
- x holds the weights
  - x_1 gives the importance of Q1 in predicting the end-term score (label)
  - x_2 gives the importance of Q2 in predicting the end-term score (label)

Find x such that Ax is as close to b as possible. Your prediction should be as close to the target as possible. You can define the error in prediction as:

$$
\begin{aligned}
e_1^2 &= (50x_1 + 70x_2 - 80)^2 \\
&\vdots \\
e_4^2 &= (100x_1 + 90x_2 - 80)^2
\end{aligned}
$$

You therefore have the SSE which you have to minimize

$$\boxed{\min_{x_1,x_2} \quad e_1^2 + \cdots + e_4^2}$$

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

You can express the predicted scores as $a_i^T x$. The SSE, often called the loss function (error function).

$$\boxed{L(x) = \left(a_1^T x - b_1\right)^2 + \; \cdots \; + \left(a_m^T x - b_m\right)^2}$$

Your goal is to:

$$\min_{x} \; L(x)$$

- $L(x)$ is called the objective function
- Unconstrained optimization (week-8)

6. Normal equations

Solution strategy:

- Differentiate
- Set derivative to zero

$L(x)$ is a function of x, where $x = (x_1, \cdots, x_n)$. Therefore, you have to compute the gradient and set it to zero.

$$\nabla L(x) = \begin{bmatrix} \dfrac{\partial L}{\partial x_1} \\ \vdots \\ \dfrac{\partial L}{\partial x_n} \end{bmatrix}$$

$$L(x) = \left(a_1^T x - b_1\right)^2 + \cdots + \left(a_m^T x - b_m\right)^2$$

$$\boxed{\nabla\left(a_i^T x - b_i\right)^2 = 2\left(a_i^T x - b_i\right)a_i}$$

$$\frac{1}{2}\nabla L(x) = \left(a_1^T x - b_1\right)a_1 + \cdots + \left(a_m^T x - b_m\right)a_m$$

$$a_1\left(a_1^T x - b_1\right) + \cdots + a_m\left(a_m^T - b_m\right)$$

$$a_1 a_1^T x - a_1 b_1 + \cdots + a_m a_m^T x - a_m b_m$$

- a_i is a vector
- x is a vector
- b_i is a scalar

$$\frac{1}{2}\nabla L(x) = \left(a_1 a_1^T + \cdots + a_m a_m^T\right)x - [b_1 a_1 + \cdots + b_m a_m] \quad (1)$$

The product of two matrices can be written as the sum of rank-1 matrices:

$$A^T = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_m \\ | & & | \end{bmatrix},\ A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

$$A^T A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_m \\ | & & | \end{bmatrix} \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

$$A^T A = a_1 a_1^T + \cdots + a_m a_m^T$$

$$A^T b = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_m \\ | & & | \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = b_1 a_1 + \cdots + b_m a_m$$

$$\frac{1}{2} \nabla L(x) = \left(A^T A\right) x - A^T b \tag{2}$$

Now setting the gradient to zero:

$$\boxed{\left(A^T A\right) x = A^T b}$$

These are the called the normal equations. Solving this will give you the solution to what is called the least squares problem. The solution is called the least squares solution. This was pioneered by Gauss in astronomy.

- This is also a system of linear equations:
  - A^TA is of shape $n \times n$
  - $x \in \mathbb{R}^n$
  - $A^Tb \in \mathbb{R}^n$
    * A^T is $n \times m$
    * $b \in \mathbb{R}^m$
- What is the guarantee that $(A^TA)x = A^Tb$ is consistent?
  - One can show that this system is always consistent
- When does $(A^TA)x = A^Tb$ have a unique solution?
  - When $\mathcal{N}(A) = \{0\}$
  - A^TA is invertible provided A has full column rank

For example:

$$\begin{bmatrix} 50 & 70 \\ 60 & 60 \\ 70 & 80 \\ 100 & 90 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \approx \begin{bmatrix} 80 \\ 60 \\ 60 \\ 80 \end{bmatrix}$$

Step-1: Find $A^T A$

Step-2: $A^T b$

Step-3: Find the augmented matrix

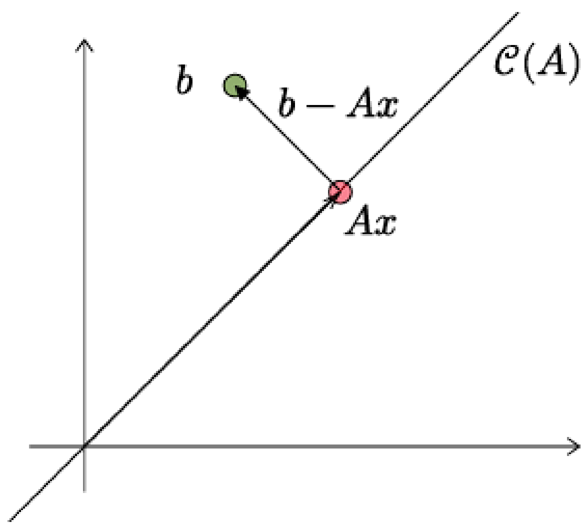
Step-4: Do row reduction and solve for x, find some particular solution

7. Projections

We are trying to find a vector Ax that is as close to b as possible. What is Ax? It is a linear combination of the columns of A. Therefore, Ax is some vector in the column space of A.

Find a vector in the column space of A that is closest to b

Recall that $\mathcal{C}(A)$ is a subspace of $\mathbb{R}^m$ and $b \in \mathbb{R}^m$



Observe that $b - Ax$ is perpendicular to the column space of A. This means $b - Ax$ is perpendicular to every column of A. The columns of A

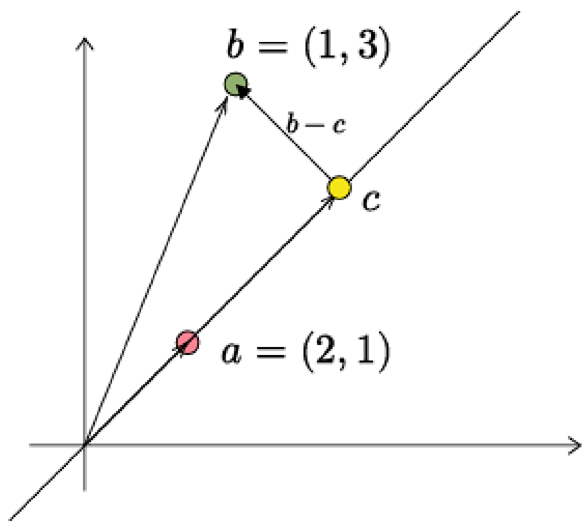
are the rows of A^T

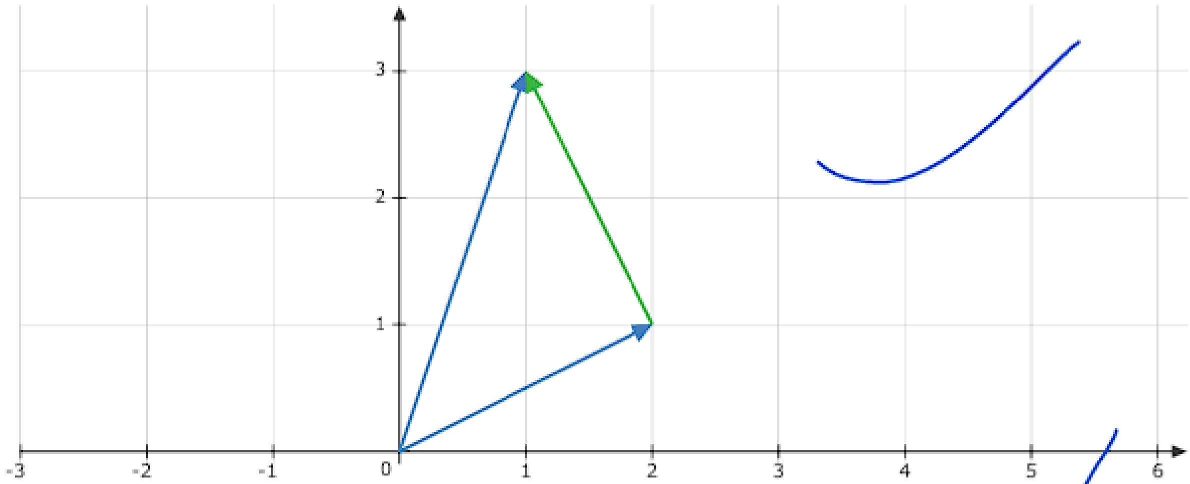
$$A^T(b - Ax) = 0$$

$$A^T b = A^T A x$$

$$\boxed{(A^T A)x = A^T b}$$

Geometrically, the least squares solution is trying to find the orthogonal projection of b onto column space of A. The resulting projection Ax.





$$a \perp (b - c)$$

$$a^T(b - c) = 0$$

$$a^T(b - \alpha a) = 0$$

Scalar projection

$$\boxed{\alpha = \frac{a^T b}{a^T a}}$$

The projection of b onto the line given by the vector a:

$$\frac{a^T b}{a^T a} \cdot a = \frac{a^T b}{||a||^2} \cdot a = \left(\frac{a}{||a||}\right)^T b \cdot \left(\frac{a}{||a||}\right)$$

$$b = (1, 3),\ a = (2, 1)$$

$$\frac{5}{5} \cdot (2, 1)$$

In general, the projection of a vector b onto a vector a is given by:

$$\boxed{\frac{b^T a}{a^T a} a}$$

If the vector a happens to have unit norm, meaning $||a||^2 = a^T a = 1$, then, the projection of b onto a can be expressed as:

$$(b^T a)a$$

8. Projection matrices

Projection of b onto the line a:

$$\frac{b^T a}{a^T a} a = a \frac{a^T b}{a^T a} = \left(\frac{1}{a^T a} \cdot a a^T \right) b$$

$$\boxed{P = \frac{a a^T}{a^T a}}$$

Therefore, the projection of b onto a is given by Pb:

$$\boxed{b \to Pb}$$

You can verify that $b - Pb \perp a$

$$a^T \left(b - \frac{a a^T b}{a^T a} \right) = a^T b - \frac{a^T a}{a^T a} \cdot a^T b = 0$$

$$P^2 = \frac{1}{\left(a^T a\right)^2} a a^T a a^T = \frac{a a^T}{a^T a} = P$$

If $a \in \mathbb{R}^n$, then P is of shape $n \times n$. P projects vectors in $\mathbb{R}^n$ onto the line which passes through the vector a.

In general, you can find a projection matrix for any subspace of $\mathbb{R}^n$. Projection matrices satisfy the following properties:

- $P^2 = P$
- P is symmetric

PA : 5,8
GA : 3,8,9,11,13

Rank nullity Theorem :

The Rank–Nullity Theorem is a fundamental result in Linear Algebra that relates:

1. **Rank of a matrix → number of linearly independent columns**
2. **Nullity of a matrix → dimension of the solution space of Ax=0Ax = 0Ax=0**
3. **Number of columns of the matrix - n**

For any matrix AAA with n columns:

Rank(A)+Nullity(A)=n\text{Rank}(A) + \text{Nullity}(A) = nRank(A)+Nullity(A)=n